MAAPC Quadcopter Report

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1 Equilibrium point

The state of the system is given by the vector $\mathbf{x} = [x, y, z, v_x, v_y, v_z, \phi, \theta, \psi, \omega_x, \omega_y, \omega_z]^T$ The control input is defined as $\mathbf{u} = [v_1^2, v_2^2, v_2^3, v_4^2]^T$

When the system is in equilibrium, the components of the derivative of the state vector are all equal to 0, as the quadcopter is not moving in any direction, nor rotating in any direction. This means that its velocities and angular velocities are all 0.

$$\dot{x} = v_x = 0 \tag{1}$$
$$\dot{y} = v_y = 0 \tag{2}$$
$$\dot{z} = v_z = 0 \tag{3}$$
$$\dot{v}_x = 0 \tag{4}$$
$$\dot{v}_y = 0 \tag{5}$$
$$\dot{v}_z = -g + \frac{k\,c_m}{m}(v_1^2 + v_2^2 + v_3^2 + v_4^2) = 0 \tag{6}$$
$$\dot{\phi} = \omega_x = 0 \tag{7}$$
$$\dot{\theta} = \omega_y = 0 \tag{8}$$
$$\dot{\psi} = \omega_z = 0 \tag{9}$$
$$\dot{\omega}_x = \frac{L\,k\,c_m}{I_{xx}}(v_1^2 - v_3^2) = 0 \tag{10}$$
$$\dot{\omega}_y = \frac{L\,k\,c_m}{I_{yy}}(v_2^2 - v_4^2) = 0 \tag{11}$$
$$\dot{\omega}_z = \frac{b\,c_m}{I_{zz}}(v_1^2 - v_2^2 + v_3^2 - v_4^2) = 0 \tag{12}$$

The equilibrium point for the inputs can be calculated using Equation (6), being that this is the only non-linear equation. This equation lets us find the values of the input voltages that guarantee that the quadcopter stays in place, which means the force being applied by the motors is the same as the force of gravity.

From equations (10), (11) and (12) we can gather that all the input voltages are the same at the equilibrium point ($v_{1_e} = v_{2_e} = v_{3_e} = v_{4_e}$), which makes since the quadcopter is at rest, which means it is not moving in any direction and so all actuating voltages have to be equal.

From this we can calculate the equilibrium point by realizing the following steps:

$$0 = -g + \frac{k \cdot cm}{m}(v_{1_e}^2 + v_{2_e}^2 + v_{3_e}^2 + v_{4_e}^2)$$
$$\frac{gm}{k \cdot cm} = v_{1_e}^2 + v_{2_e}^2 + v_{3_e}^2 + v_{4_e}^2$$
$$v_{1_e}^2 = v_{2_e}^2 = v_{3_e}^2 = v_{4_e}^2 = 40.8750$$

From this we can conclude that the input vector u at the equilibrium point $U_e = \begin{bmatrix} 40.8750 & 40.8750 & 40.8750 & 40.8750 \end{bmatrix}^T$

2 Linearization

With the equilibrium point found we can now linearize the system around it:

$$\Delta\dot{\mathbf{x}} = A\,\Delta\mathbf{x} + B\,\Delta\mathbf{u} \tag{13}$$

$$\Delta\dot{x} = A\begin{bmatrix}\Delta\dot{x}\\ \Delta\dot{y}\\ \Delta\dot{z}\\ \Delta\dot{v}_x\\ \Delta\dot{v}_y\\ \Delta\dot{v}_z\\ \Delta\dot{\phi}\\ \Delta\dot{\theta}\\ \Delta\dot{\psi}\\ \Delta\dot{\omega}_x\\ \Delta\dot{\omega}_y\\ \Delta\dot{\omega}_z\end{bmatrix} + B\begin{bmatrix}\Delta v_1\\ \Delta v_2\\ \Delta v_3\\ \Delta v_4\end{bmatrix} = A\begin{bmatrix}\Delta x_1\\ \Delta x_2\\ \Delta x_3\\ \Delta x_4\\ \Delta x_5\\ \Delta x_6\\ \Delta x_7\\ \Delta x_8\\ \Delta x_9\\ \Delta x_{10}\\ \Delta x_{11}\\ \Delta x_{12}\end{bmatrix} + B\begin{bmatrix}\Delta u_1\\ \Delta u_2\\ \Delta u_3\\ \Delta u_4\end{bmatrix} \tag{14}$$

After changing the states names to something more practical we could start the linearization:

$$\dot{x}_1 = x_4 \tag{15}$$
$$\dot{x}_2 = x_5 \tag{16}$$
$$\dot{x}_3 = x_6 \tag{17}$$
$$\dot{x}_4 = -\frac{k_d}{m}x_4 + \frac{kc_m}{m}\left(\sin(x_9)\sin(x_7) + \cos(x_9)\cos(x_7)\sin(x_8)\right)(u_1+u_2+u_3+u_4) \tag{18}$$
$$\dot{x}_5 = -\frac{k_d}{m}x_5 + \frac{kc_m}{m}(\cos(x_7)\sin(x_9)\sin(x_8) - \cos(x_9)\sin(x_7))(u_1+u_2+u_3+u_4) \tag{19}$$
$$\dot{x}_6 = -\frac{k_d}{m}x_6 - g + \frac{kcm}{m}\cos(x_8)\cos(x_7)(u_1+u_2+u_3+u_4) \tag{20}$$
$$\dot{x}_7 = x_{10} + x_{11}\sin(x_7)\tan(x_8) + x_{12}\cos(x_7)\tan(x_8) \tag{21}$$
$$\dot{x}_8 = x_{11}\cos(x_7) - x_{12}\sin(x_7) \tag{22}$$
$$\dot{x}_9 = \frac{\sin(x_7)}{\cos(x_8)}x_{11} + \frac{\cos(x_7)}{\cos(x_8)}x_{12} \tag{23}$$
$$\dot{x}_{10} = \frac{Lkc_m}{I_{xx}}(u_1-u_3) - \frac{I_{yy}-I_{zz}}{I_{xx}}x_{11}x_{12} \tag{24}$$
$$\dot{x}_{11} = \frac{Lkc_m}{I_{yy}}(u_2-u_4) - \frac{I_{zz}-I_{xx}}{I_{yy}}x_{10}x_{12} \tag{25}$$
$$\dot{x}_{12} = \frac{Lkc_m}{I_{zz}}(u_1-u_2+u_3-u_4) - 0 \tag{26}$$

$$\begin{aligned} x_e &= 0 & x &= \Delta x\\ y_e &= 0 & x &= \Delta y\\ u_e &= 40.8750 & u &= \Delta u + 40.875 \end{aligned}$$

Next we can work on the linear approximation equations. These are obtained by taking the first order terms of the Taylor Series Expansion:

$$\Delta\dot{x}_1 = \Delta x_4 \tag{27}$$

$$\Delta\dot{x}_2 = \Delta x_5 \tag{28}$$

$$\Delta\dot{x}_3 = \Delta x_6 \tag{29}$$

$$\Delta\dot{x}_4 = -\frac{k_d}{m}\Delta x_4 + (163.5)\frac{kc_m}{m}\Delta x_8 \tag{30}$$

$$\Delta\dot{x}_5 = -\frac{k_d}{m}\Delta x_5 - \frac{kc_m}{m}(163.5)\Delta x_7 \tag{31}$$

$$\Delta\dot{x}_6 = -\frac{k_d}{m}\Delta x_6 + \frac{kc_m}{m}(\Delta u_1 + \Delta u_2 + \Delta u_3 + \Delta u_4) \tag{32}$$

$$\Delta\dot{x}_7 = \Delta x_{10} \tag{33}$$

$$\Delta\dot{x}_8 = \Delta x_{11} \tag{34}$$

$$\Delta\dot{x}_9 = \Delta x_{12} \tag{35}$$

$$\Delta\dot{x}_{10} = \frac{Lkc_m}{I_{xx}}(\Delta u_1 - \Delta u_3) \tag{36}$$

$$\Delta\dot{x}_{11} = \frac{Lkc_m}{I_{yy}}(\Delta u_2 - \Delta u_4) \tag{37}$$

$$\Delta\dot{x}_{12} = \frac{bc_m}{I_{zz}}(\Delta u_1 - \Delta u_2 + \Delta u_3 - \Delta u_4) \tag{38}$$

With these equations we can extract the state space model for the linearized system.

$$\Delta\dot{x} = A\Delta x + B\Delta u$$
$$\Delta\dot{y} = C\Delta x$$

To make sparse matrices easier to read, we are going to use the notation $\cdot$ for cells including zeros in the matrix.

$$A = \begin{bmatrix}
\cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & -\frac{k_d}{m} & \cdot & \cdot & \cdot & 163.5\frac{kc_m}{m} & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & -\frac{k_d}{m} & \cdot & -163.5\frac{kc_m}{m} & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & -\frac{k_d}{m} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot
\end{bmatrix}$$

$$B = \begin{bmatrix}
\cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot \\
\frac{kc_m}{m} & \frac{kc_m}{m} & \frac{kc_m}{m} & \frac{kc_m}{m} \\
\cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot \\
\frac{Lkc_m}{I_{xx}} & \cdot & -\frac{Lkc_m}{I_{xx}} & \cdot \\
\cdot & \frac{Lkc_m}{I_{yy}} & \cdot & -\frac{Lkc_m}{I_{yy}} \\
\frac{bc_m}{I_{zz}} & -\frac{bc_m}{I_{zz}} & \frac{bc_m}{I_{zz}} & -\frac{bc_m}{I_{zz}}
\end{bmatrix}$$

$$C = \begin{bmatrix} 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot \end{bmatrix}$$

When comparing the nonlinear model's to the linearized model's step response, the following results were obtained:

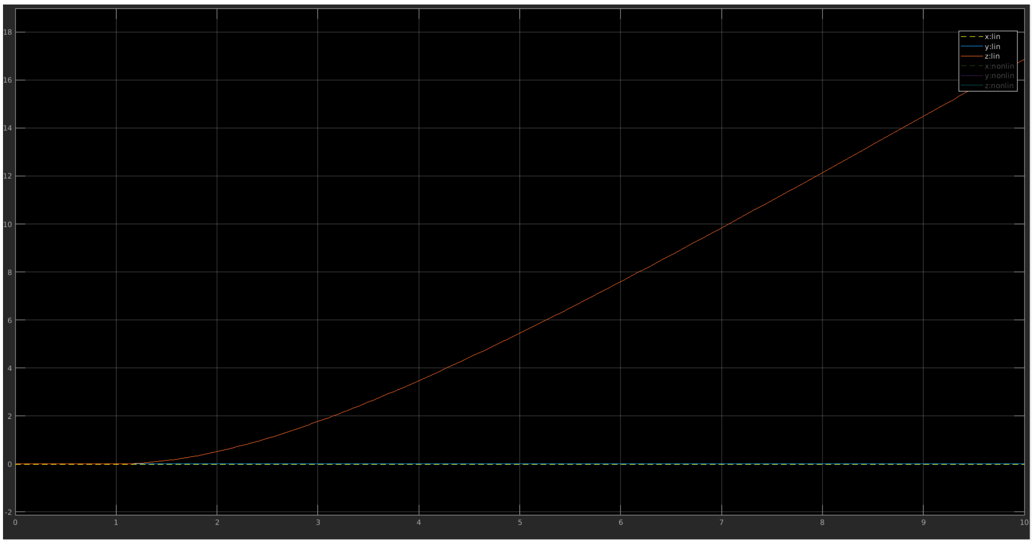

Figure 1: Step response of linear system position (x, y, z).

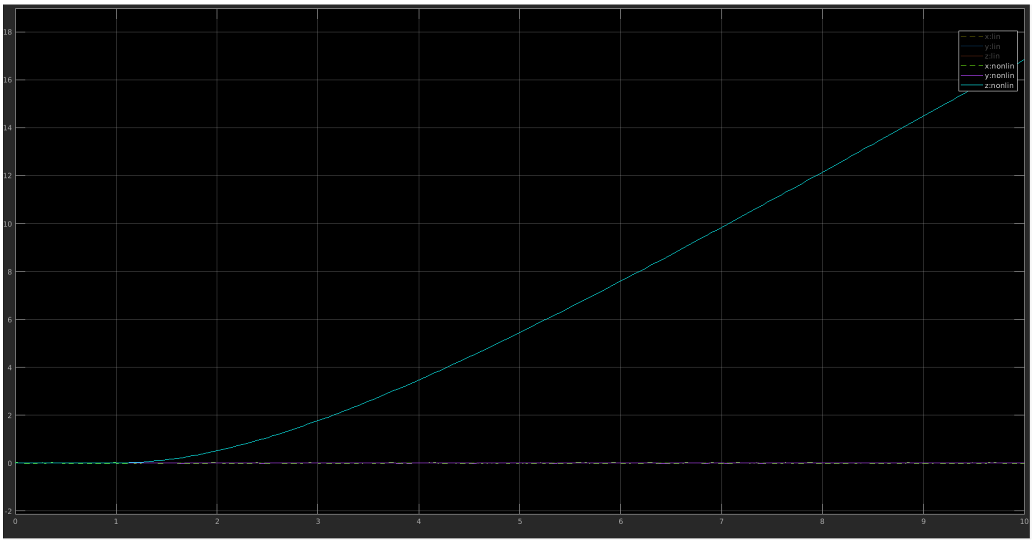

Figure 2: Step response of non-linear system position (x, y, z).

Being that the linearized model performs similarly to the non-linear model, we can conclude that it is a good approximation for this system's behaviour around the equilibrium point.

3 Discretization

For the discretization of the model we chose ZOH. The new system can be defined as follows:

From checking the ranks of the controllability and observability matrices we can conclude that the system is both controllable and observable, which means it is also sterilizable, detectable and therefore minimal. By using the command *tzero* we gathered that the system has 2 transmission zeros:

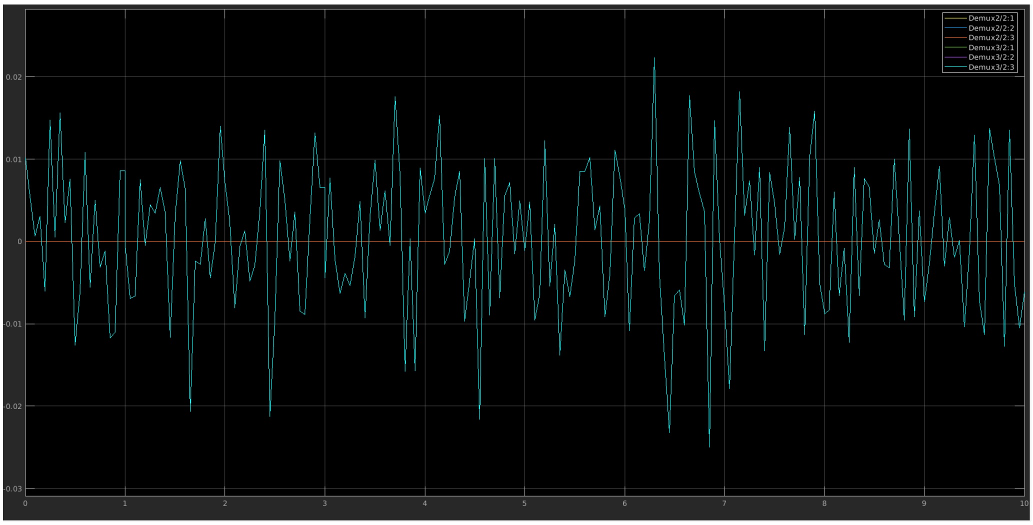

Figure 3: Step response of both systems angles (ϕ, θ, ψ). All angles of linear system are at the straight zero line and all the signals from non-linear system covers the noisy one.

- $z = (-0.9917)$
- $z = (-1.0000)$

The system has a sextuple pole at $z = 1$, which means it is marginally stable.

4 LQR control

Experiments without payload

Our initial design assumes that errors in positions should be slightly more penalized than the one related to angles, because we need to perturb angles in order to move the quadcopter between positions and our main objective is to reach reference points.

```
Q = diag([2, 2, 2, 0, 0, 0, 1, 1, 1, 0, 0, 0])
R = eye(4)
```

As the dynamics of the system is much slower in case of z-axis than other, and quadcopter have most problems with tracking height, we increased the penalization factor related to z coordinate.

```
Q = diag([2, 2, 20, 0, 0, 0, 2, 2, 0.1, 0, 0, 0])
R = eye(4)
```

The controller was still not able to reach all the reference points, we decided to make it more stable. To do so, we penalized ϕ and θ angles. We found ψ angle not as important for the controller design, as it is not related to keeping the drone vertical.

```
Q = diag([2, 2, 20, 0, 0, 0, 2, 2, 0.1, 0, 0, 0])
R = eye(4)
```

Then we penalized velocities and angular velocities of quadcopter. Penalization of only the angular velocities decreased our performance, but with penalization of velocities, the time of run remains this same as before. We aimed to make the quadcopter more robust by preventing high angular velocities. We end up with a controller. Tracking the control actions during the design process, we decided to stay with unit values on R matrix. The values of control actions are reasonably high, but there are no over-saturations of motors.

```
Q = diag([2, 2, 20, 1, 1, 1, 2, 2, 0.1, 0.1, 0.1, 0.1])
R = eye(4)
```

Tracking behaviour is implemented with

$$u = -K(x - N_x) + N_u$$

where N_x is equal to reference points concatenated with zeros and $N_u = [u_e, u_e, u_e, u_e]$.

LQR with integral action

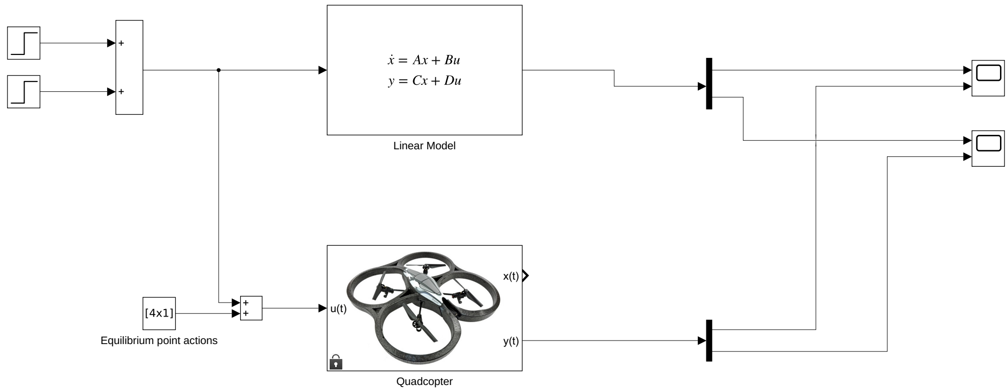

Figure 4: Simulink diagram used to compare linearization to the original system.

To incorporate integral action to LQR design, we augmented the system with additional states. The additional states correspond to accumulated errors on x, y and z positions. In the result, we got two matrices NA and NB, which could be used to compute the control law K for the augmented system. The system constructed with NA and NB matrices is controllable, so we can design the controller. However, to use the control law for the provided system, we had to split it into two gains K_s and K_i.

The Q matrix of new control gained 3 additional parameters. Each of them corresponds to an integration term. Too high values of these parameters are leading to overshoot the reference. In other hand, too low values may not lead the system to reference point fast enough. After few trials, we found that z coordinate of position needs a little bit greater integration term than x and y. These observations made us develop the following controller parameters.

```
Q = diag([0.2, 0.2, 0.5, 2, 2, 20, 1, 1, 1, 2, 2, 0.1, 0.1, 0.1, 0.1])
R = eye(4)
```

Experiment with payload In case of full-state feedback controller with no integral actions, the control law can't detect change of weight. This difference in weight causes the control action N_u not lead to equilibrium state. The simplest explanation why it was not able to control the system successfully is that the model changed, and it was different from the one for which the controller was designed.

The LQR with integral actions was robust enough to handle the run with a 0.1 kg payload. The result of adding payload was the decrease in the speed of the flight from average time between checkpoints with value 5.171s to 5.764s.

The essential difference between both control designs is that the LQR with integral actions was able to work properly even with system which differ from the one it was designed for. The first design is much more sensitive to model misalignments.

5 LQG Control

The Kalman filter which extends current LQR design is implemented in a following way. We used provided variances to construct R_k matrix, which describes the sensor accuracy. To create, Q_k we have chosen to use values in similar range of magnitude and check accuracy of the output signal, comparing it with states of the system. We found that approximation for most of the states is quite accurate. The filter was inaccurate only with z and v_z estimation. We decided to increase the weight corresponding with v_z in the Q_k as long as the

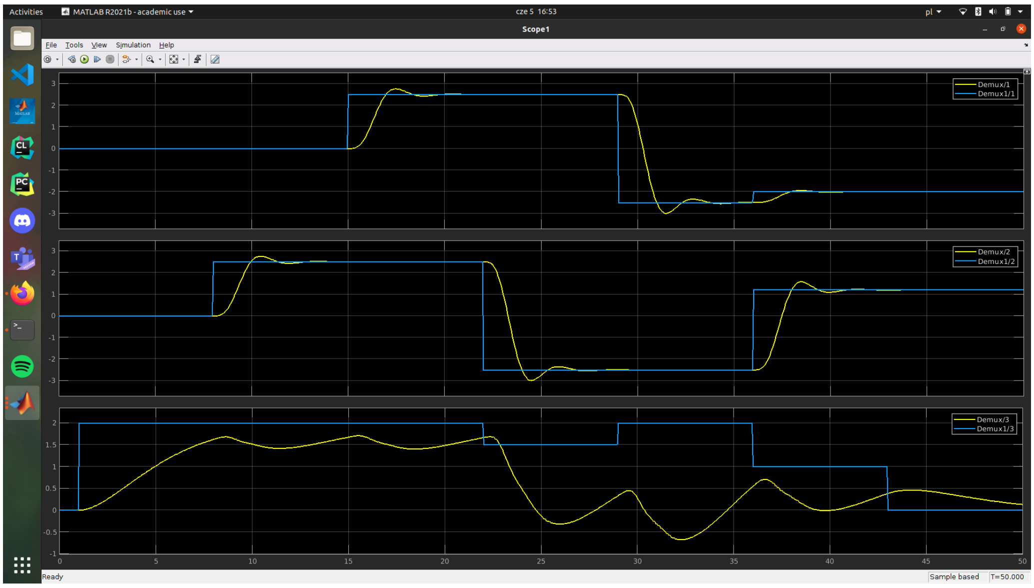

Figure 5: Difference between positions and reference points.

resulting output signal gets more accurate and is still not too noisy. We found that value $2 * 10^-4$ works fine. The improvement in the v_z estimation increased also the accuracy of z to the satisfactory level.

```
pos_var = 2.5e-5
ang_var = 7.57e-5

qk_diag = repmat(1e-5, [1, 12])
qk_diag(6) = 2e-4
Qk = diag(qk_diag)

Rk = diag([repmat(pos_var, [1, 3]), repmat(ang_var, [1, 3])])

% Kalman filter
B1 = eye(12, 12)
[M, P] = dlqe(Ad, B1, C, Qk, Rk)
L = Ad * M
```

The observer submodule is implemented as a discrete state-space model with parameters:

$$A := A_d - LC_d$$
$$B := [B_d : L]$$
$$C := I_{12\times 12} \qquad \text{(identity matrix } 12\times 12\text{)}$$
$$D := 0_{12\times 10} \qquad \text{(matrix of zeros } 12\times 10\text{)}$$

where A_d, B_d and C_d are respectively A, b and C matrices of the discrete linearized model of the quadcopter.

Experiment with no payload The LQG controller was able to finish the run, with average time of 5.557s between checkpoints.

Experiment with 0.1 kg payload With the 0.1 kg of payload, the LQG controller was not able to reach the first checkpoint. The average time for reaching the distance between the rest of checkpoints was 6.2s. The payload increased the gap between the estimated velocity in the z axis and the real one. We believe it is because of additional mass increased constant force that is pulling the quadcopter downwards.

6 Control with Pole Placement

Being that this system is a MIMO system we must use Sylvester's equation to place the poles. First we create a matrix V whose eigenvalues are the desired poles of the system. Then we create an arbitrary G matrix and solve the following equation for X using the *sylvester* method in matlab:

$$AX - XV = BG \tag{39}$$

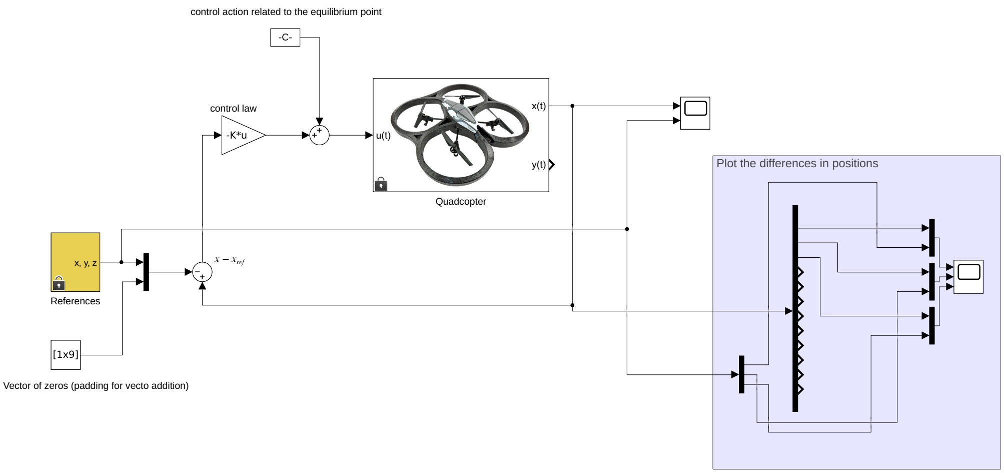

Figure 6: Simulink diagram for full-state feedback LQR.

The end result is the transformation matrix X, which we can then use to get the K matrix for the filter: $K = GX^{-1}$.

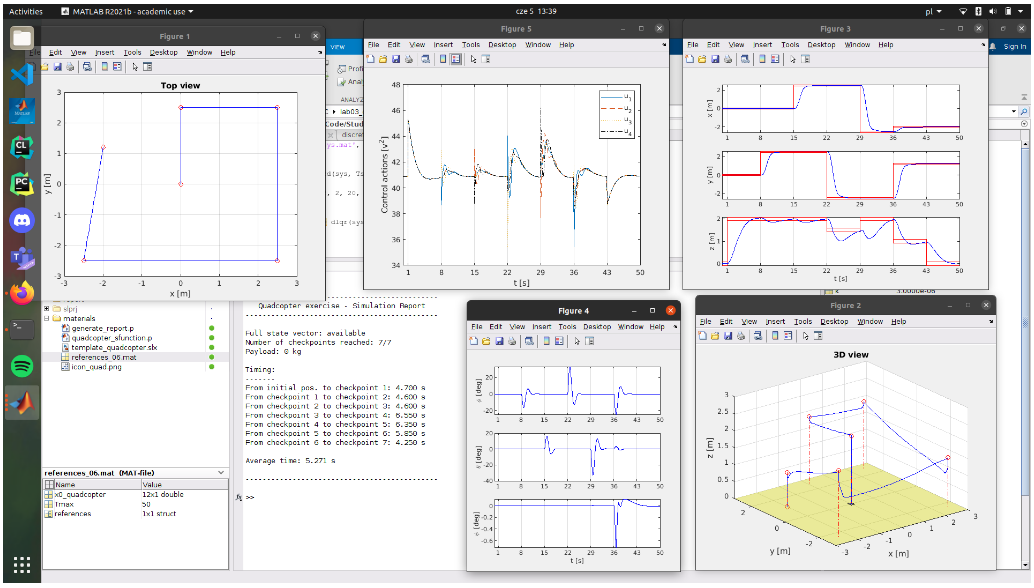

Figure 7: Report from LQR run with no payload.

```
NA = [
    eye(3, 3), eye(3, 12);
    zeros(12, 3), sys_dt.A
]
NB = [zeros(3, 4); sys_dt.B]

CM = ctrb(NA, NB);
assert(rank(CM) == 15)

Q = diag([0.2, 0.2, 0.5, 2, 2, 20, 1, 1, 1, 2, 2, 0.1, 0.1, 0.1, 0.1])
R = eye(4)

[K, S, e] = dlqr(NA, NB, Q, R, zeros(15, 4))

Ki = K(:,1:3)
Ks = K(:,4:end)
```

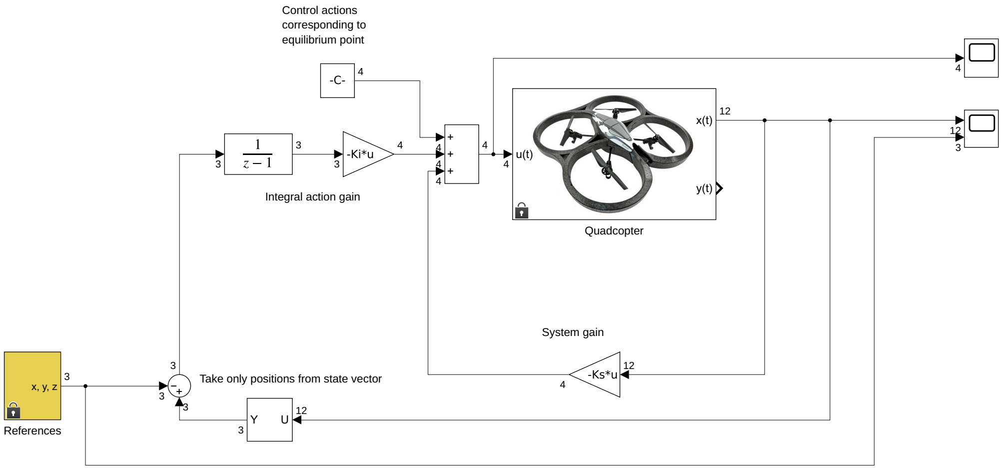

Figure 8: Simulink diagram for LQR with integral actions.

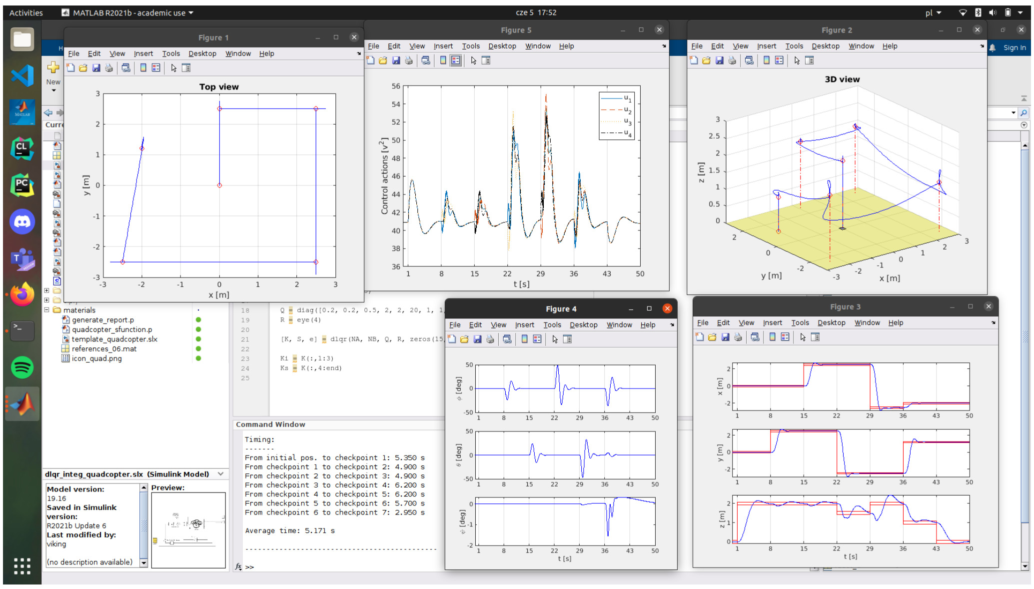

Figure 9: Report from run with no payload of LQR with integral actions.

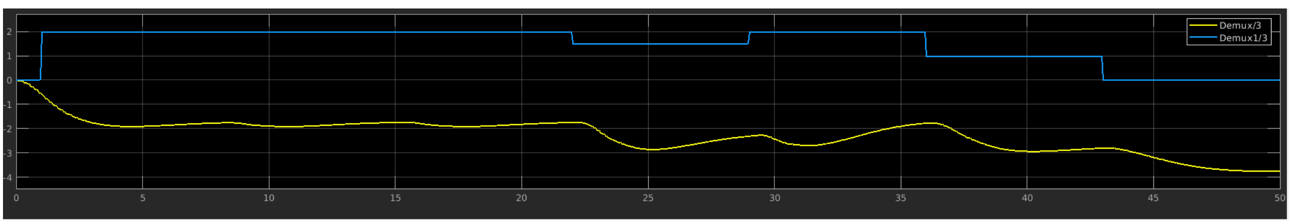

Figure 10: The z position of the quadcopter with the full-state feedback LQR controller, under payload.

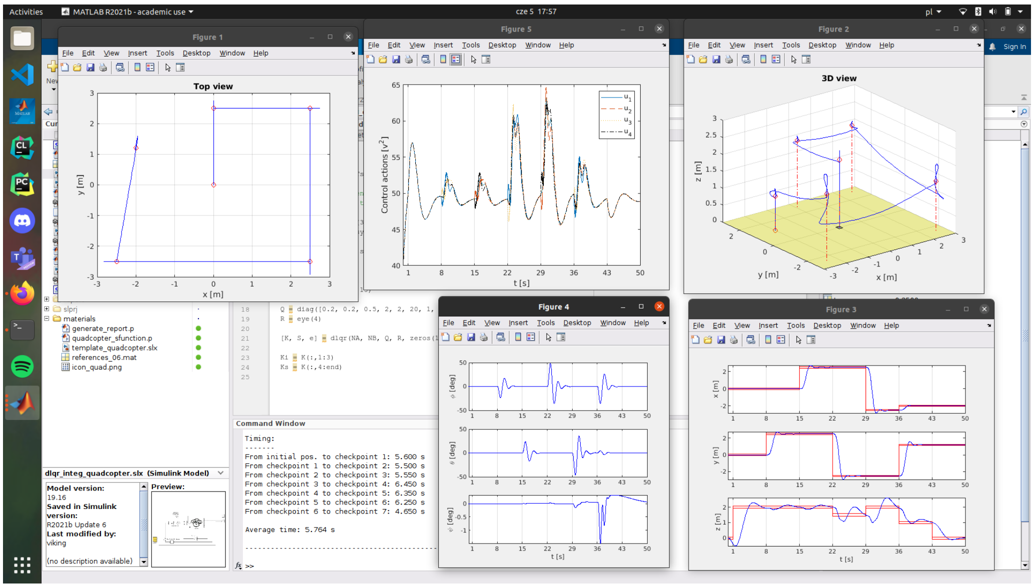

Figure 11: Report from run with 0.1 kg payload of LQR with integral actions.

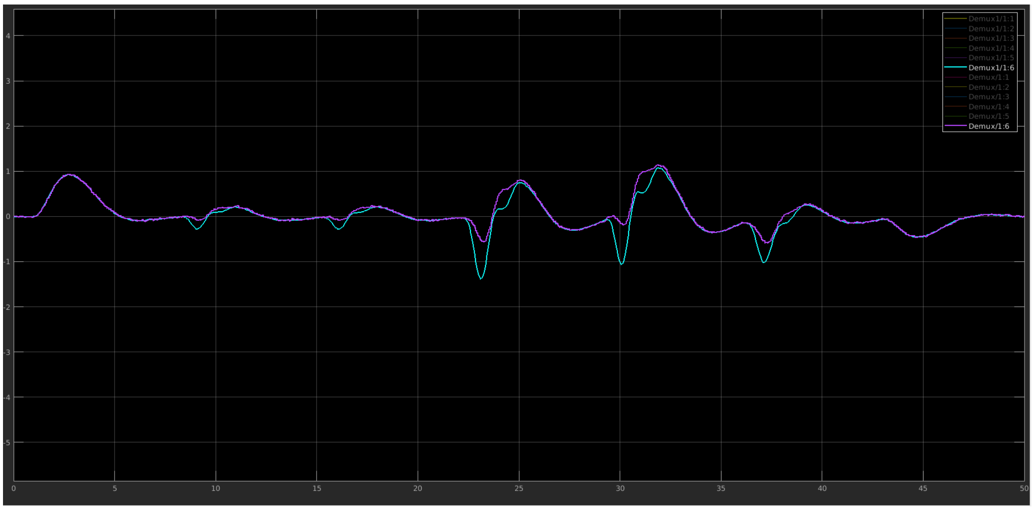

Figure 12: Estimation of v_z with Kalman filter.

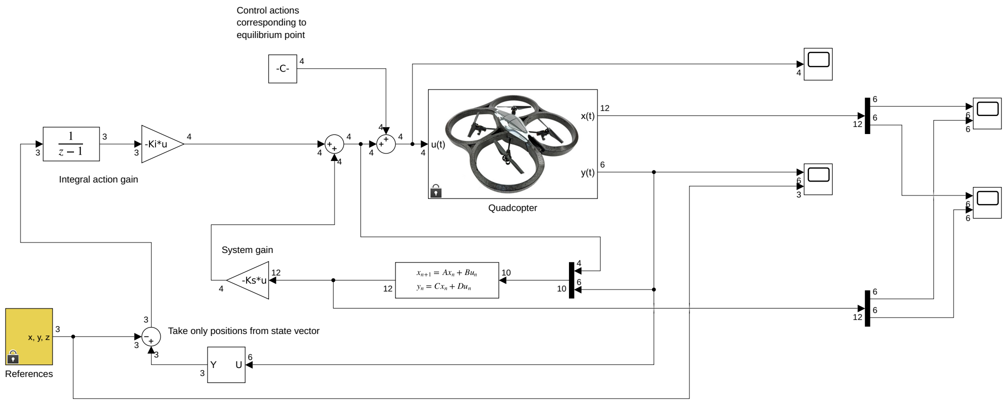

Figure 13: Simulink diagram of LQG controller.

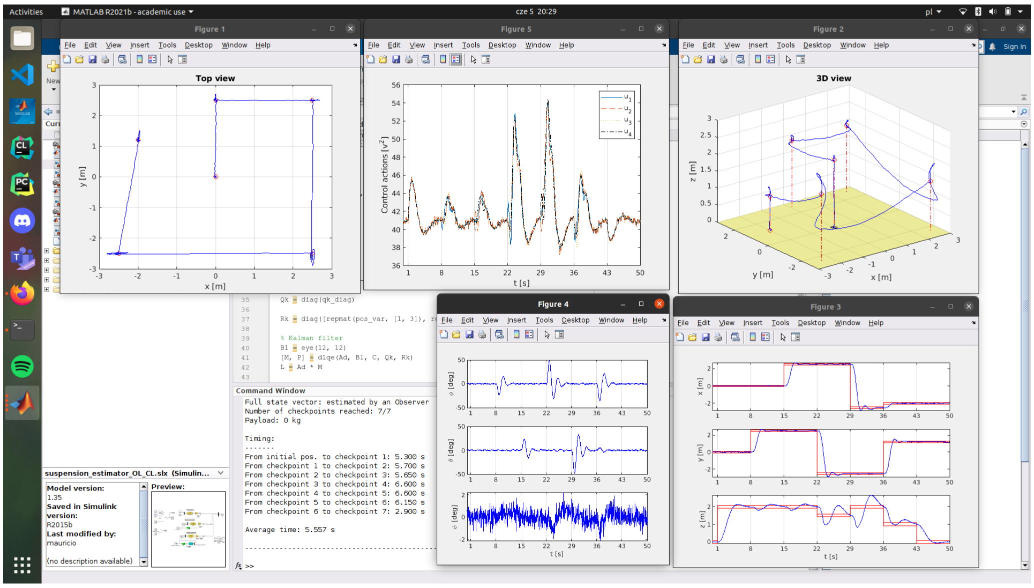

Figure 14: Report from run of LQG with no payload

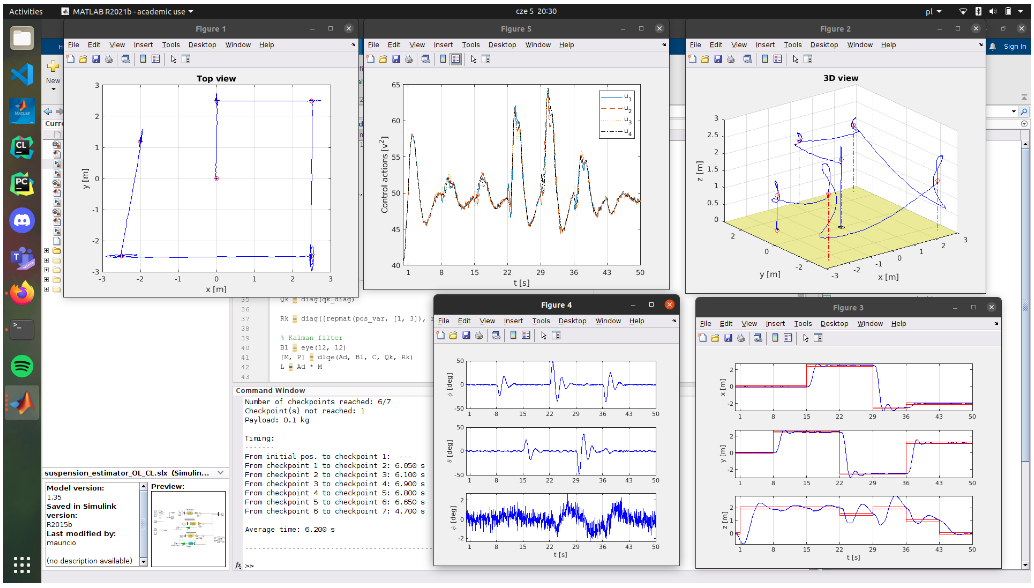

Figure 15: Report from run of LQG with 0.1 kg payload